



VMware and Private Cloud Foundation (VCF) Program

Build the Skills Behind Modern Data Centers, Private Cloud and Virtualization

SMART MATE VENTURES

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About SMART MATE VENTURES

Our Mission

Empower IT professionals with enterprise-grade VMware and private cloud skills through hands-on, industry-aligned training.

Our Vision

To be the most trusted destination for virtualization and cloud infrastructure education in the enterprise learning space.

Our Value

We don't just teach technology—we build confidence, practical skills, problem-solving abilities, and successful IT careers.

Enterprise Learning

Curriculum aligned with real-world enterprise environments

Real Lab Experience

Hands-on labs using live VMware infrastructure

Industry Trainers

Learn from certified VMware practitioners

Career Guidance

End-to-end support from enrollment to placement



Why Choose **This Program**

A comprehensive, enterprise-focused curriculum designed to take you from fundamentals to production-ready skills.



Enterprise Labs

Live, nested VMware environments



Cloud Fundamentals

Private cloud and VCF architecture



Storage

VMFS, vSAN, iSCSI, NFS



Career Guidance

Resume, LinkedIn, and job search support



Virtualization

Deep-dive into vSphere, ESXi, vCenter



Networking

vSS, vDS, NSX essentials



Certification

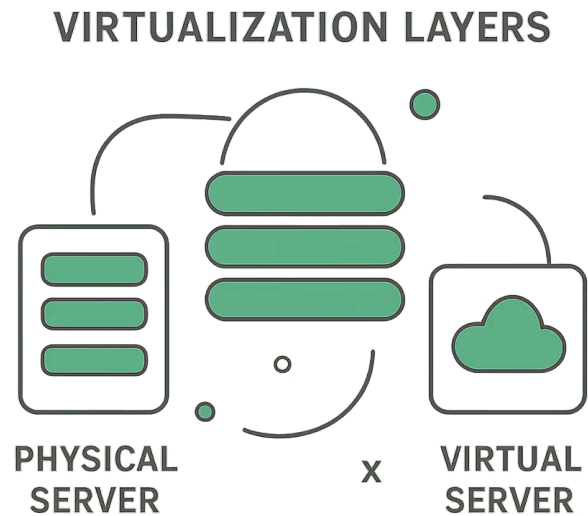
Exam-ready preparation included



Interview Prep

Mock interviews with industry scenarios

Virtualization Fundamentals



Core Concepts

- What is Virtualization & why enterprises adopt it
- Physical vs Virtual Infrastructure
- Type 1 vs Type 2 Hypervisors
- Compute, Storage & Network Virtualization
- Introduction to Private Cloud



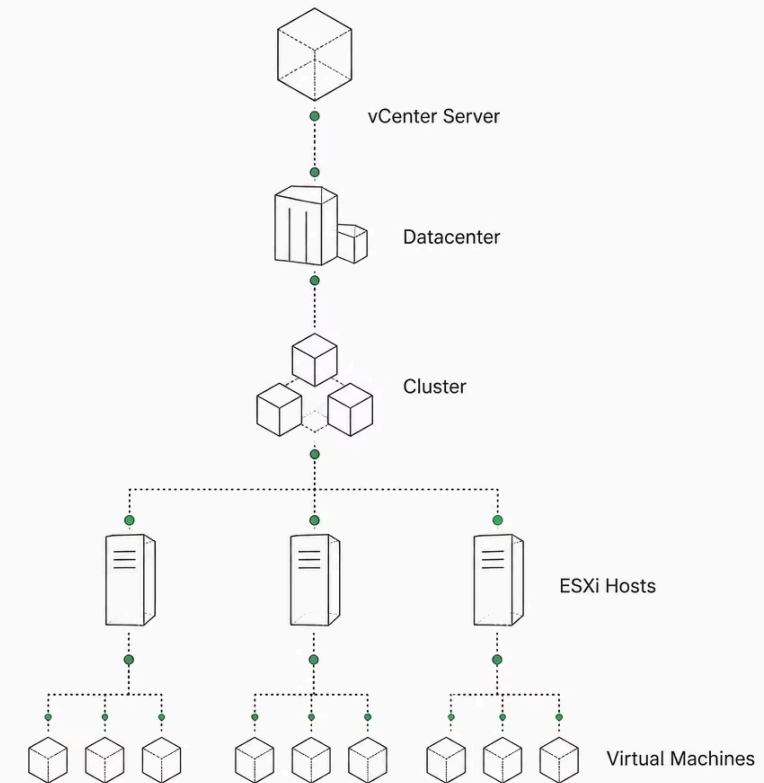
Hands-on Lab: Install VMware Workstation · Create Windows & Ubuntu VMs · Compare Physical vs Virtual Resources

VMware vSphere Architecture

Architecture Deep Dive

- VMware Product Portfolio overview
- ESXi Architecture & vCenter Server
- Datacenter, Cluster, Host & Resource Pools
- Inventory Objects & Licensing

✓  **Hands-on Lab:** Deploy ESXi · Install VCSA · Add Host to vCenter · Create Datacenter & Cluster



MODULE 3

ESXi Administration

Installation

ESXi setup, DCUI navigation,
Licensing

Configuration

NTP, DNS, Host Profiles,
SSH, Lockdown Mode

Security

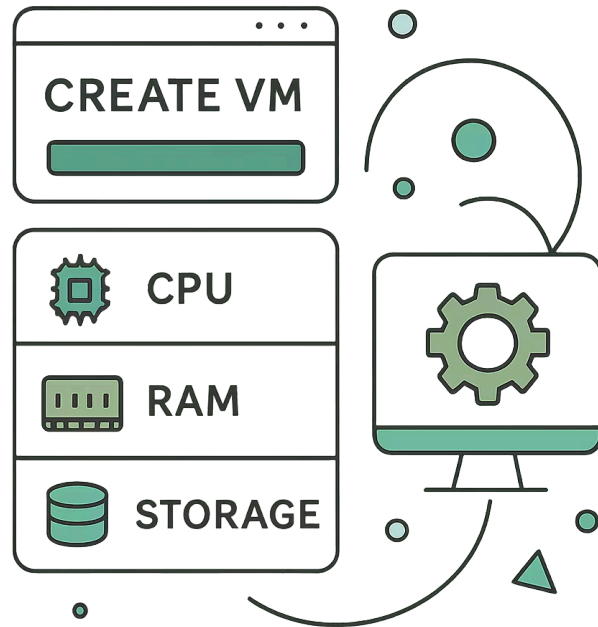
Time Synchronization, Management Network hardening



Hands-on Lab: Install ESXi · Configure Management
Network · Enable SSH · Configure NTP & DNS



Virtual Machine Administration



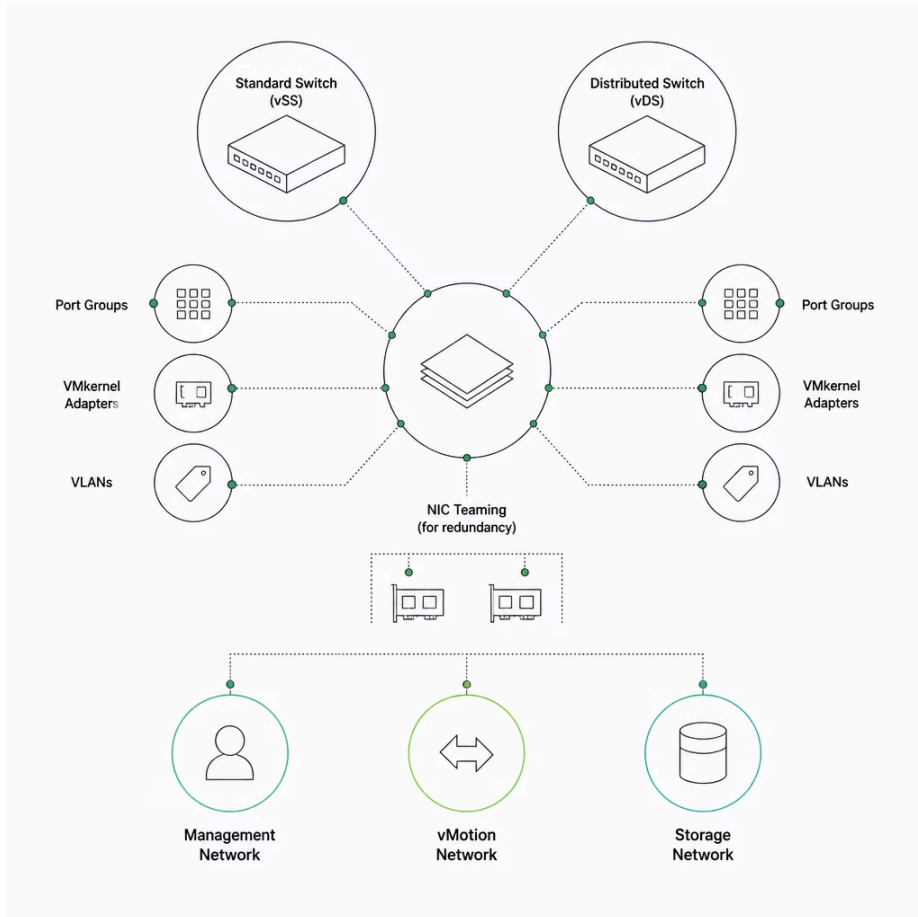
VM Lifecycle Management

- VM Creation, Hardware Versions & VMware Tools
- Snapshots, Templates & Cloning
- Guest Customization & Resource Allocation
- Hot Add & Virtual Hardware management



Hands-on Lab: Create Windows & Linux VMs · Install VMware Tools · Clone VM · Create Snapshot · Convert to Template

VMware Networking



Networking Stack

- Standard Switch (vSS) vs Distributed Switch (vDS)
- Port Groups & VMkernel Adapters
- VLANs, NIC Teaming & Load Balancing
- Network Redundancy & Failover
- vMotion Network & Storage Network



Hands-on Lab: Create vSwitch · Configure Port Groups & VMkernel · Configure vDS · Test NIC Failover

VMware Storage

Storage Protocols

- SAN & NAS
- Fibre Channel
- iSCSI
- SMB & NFS

Datastores

- VMFS
- vVols
- Thin Provisioning
- Thick Provisioning

Policies

- Storage Policies
- Storage Multipathing



Hands-on Lab: Add iSCSI Storage · Mount NFS · Create VMFS Datastore · Configure Multipathing



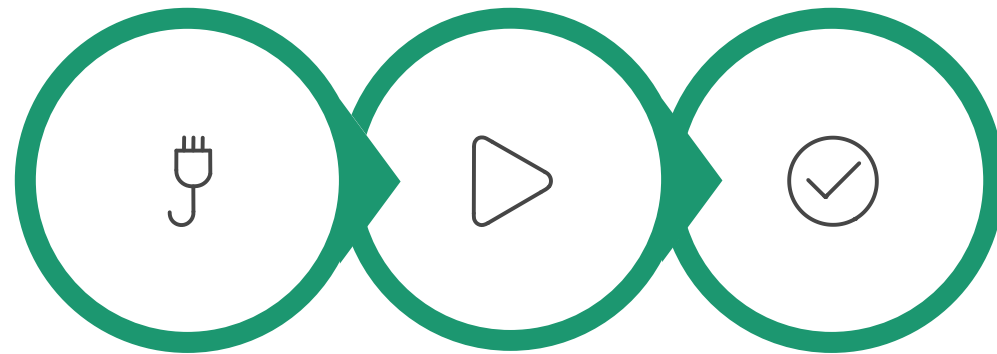
vMotion & Storage vMotion

Live Migration Capabilities

- Live VM migration with zero downtime
- Storage Migration across datastores
- Cross-vCenter vMotion
- Enhanced vMotion — prerequisites & limitations



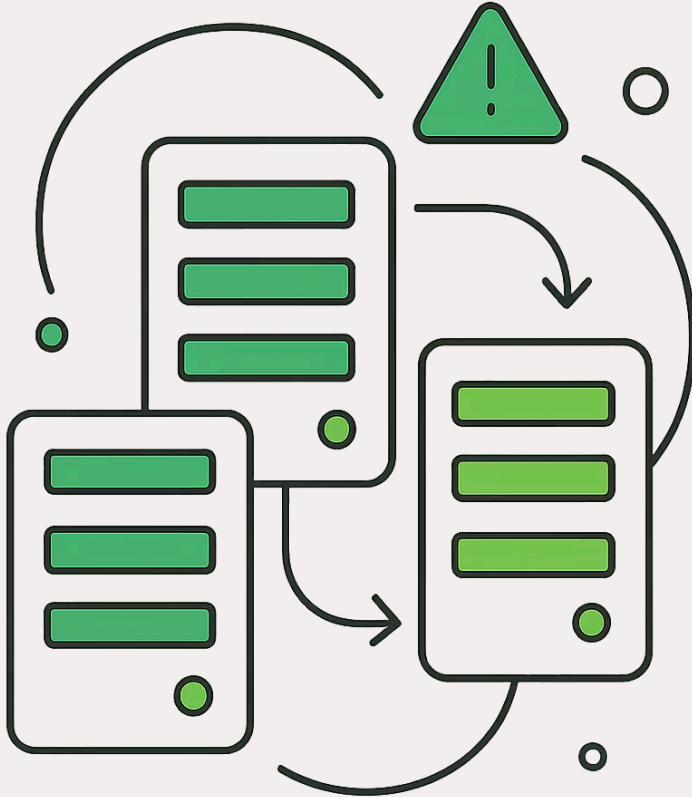
Hands-on Lab: Configure vMotion Network · Perform Live Migration · Perform Storage vMotion



Configure
vMotion

Initiate
Migration

Validate VM



MODULE 8

Cluster Features: HA & DRS

1

VMware HA

Automatic VM restart on host failure.
Configure Admission Control for resource reservation.

2

VMware DRS

Automated workload balancing across cluster hosts. Fully automated or manual recommendations.

3

EVC & Resource Pools

CPU compatibility across mixed hardware. Hierarchical resource management.



Hands-on Lab: Create Cluster · Configure HA & DRS · Test Host Failure · Host Maintenance Mode · Simulate HA Failover

VMware vSAN



**HYPER-CONVERGED
STORAGE**

Software-Defined Storage

- vSAN Architecture — ESA vs OSA
- Disk Groups & Storage Policies
- Fault Domains & Stretched Cluster
- Deduplication, Compression & Erasure Coding



Hands-on Lab: Configure vSAN Cluster · Create Disk Groups · Set Storage Policy · Deploy VM on vSAN · Test Storage Failures

MODULE 10

Monitoring & Troubleshooting

Performance

Performance Charts, esxtop, resource contention analysis

Events & Alarms

Configuring alerts, reviewing vCenter & host logs

Troubleshooting

VM power-on failures, storage connectivity issues, systematic methodology, ESXi and vCenter Log Analysis



Hands-on Lab: Configure Alarms · Analyze Performance · Troubleshoot VM Power-On · Diagnose Storage Issues



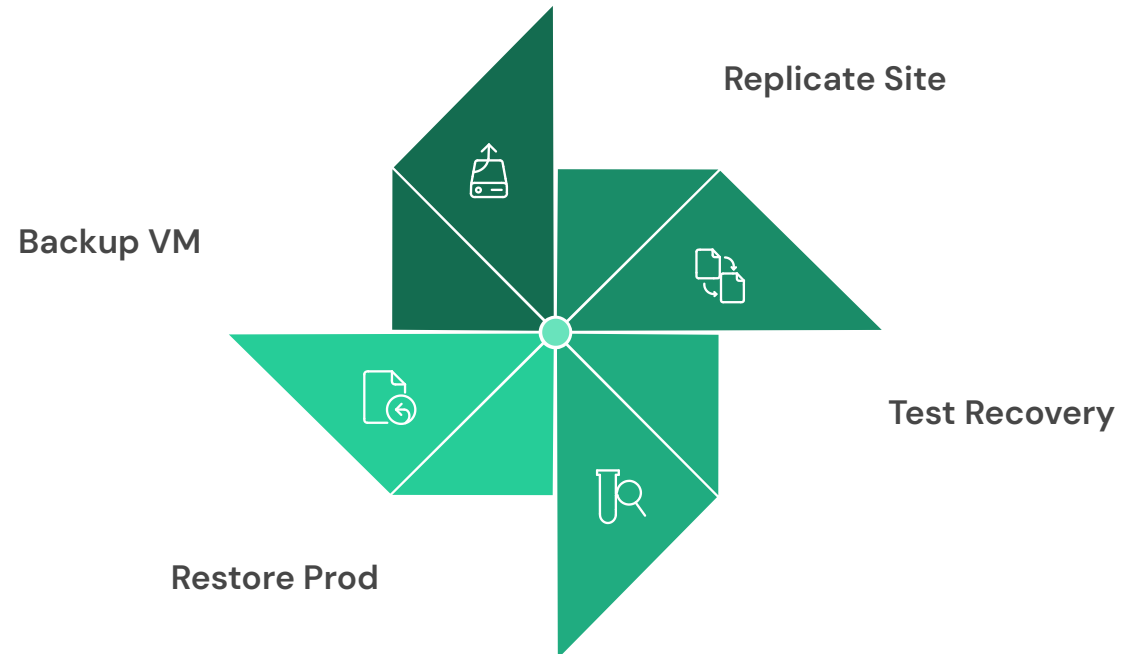
Backup & Disaster Recovery

Resilience Strategies

- VMware Backup Strategies — Snapshots vs Backup
- Replication & Site Recovery concepts
- VMware Live Recovery overview
- Veeam Integration



Hands-on Lab: Perform
VM Backup (Demo) · Restore
VM · Test Replication



VMware NSX Fundamentals



Network Virtualization

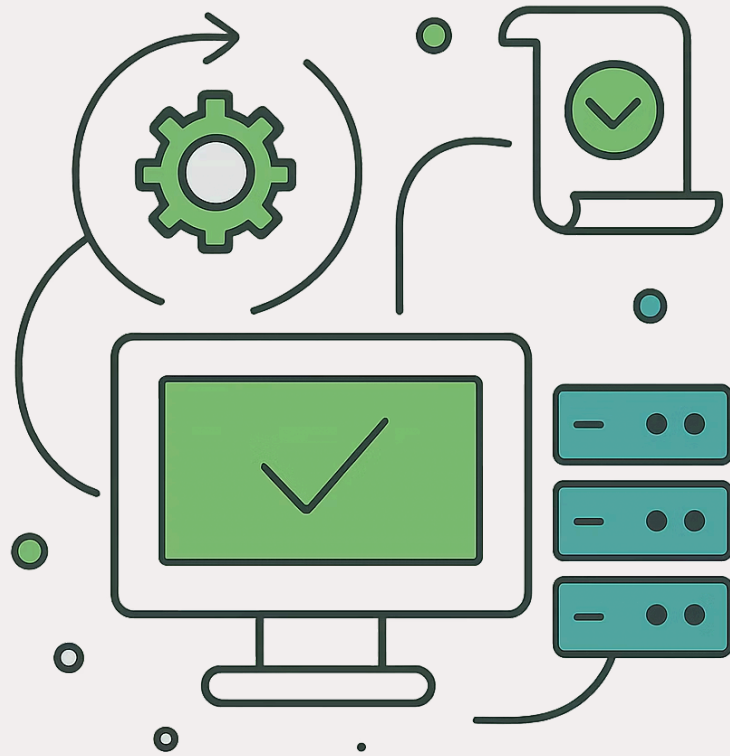
- NSX Architecture & Overlay Networks
- Logical Switching & Logical Routing
- Distributed Firewall & Micro-segmentation
- Edge Gateway & Load Balancing



Hands-on Lab: Deploy NSX Manager (Conceptual/Nested Lab) · Create Overlay Segment

MODULE 13

Lifecycle Management



1

Baseline

Create image-based update baselines

2

Scan

Compliance scan across cluster hosts

3

Remediate

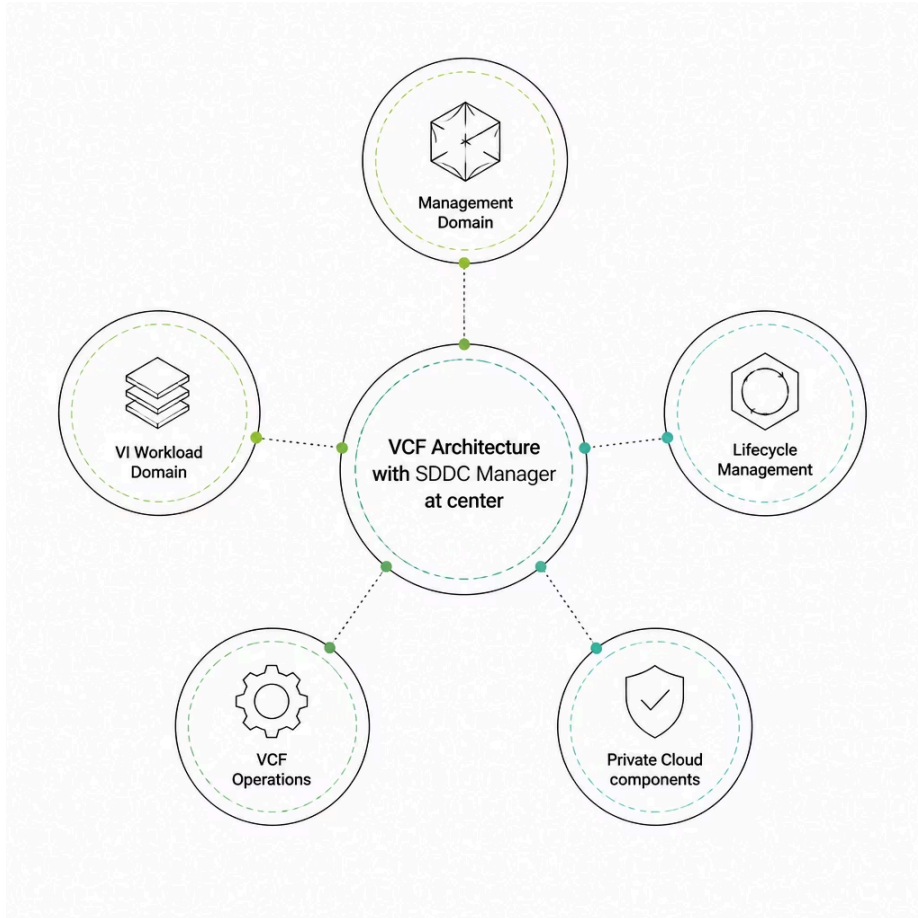
Patch & firmware remediation at scale

Covers Lifecycle Manager, Image-Based Updates, Firmware Management, Certificate Management & Patch Compliance.



Hands-on Lab: Create Baseline · Scan Cluster · Remediate ESXi Hosts

VMware Cloud Foundation (VCF)



Private Cloud at Scale

- What is VCF? — Versions & Architecture
- SDDC Manager & Workload Domains
- Management Domain vs VI Workload Domain
- Bill of Materials, VCF Components & Operations



Hands-on Lab: Explore VCF Architecture · Navigate SDDC Manager · Review Lifecycle Management

Enterprise Design & Best Practices

HA Design

High Availability architecture patterns for production environments

Capacity Planning

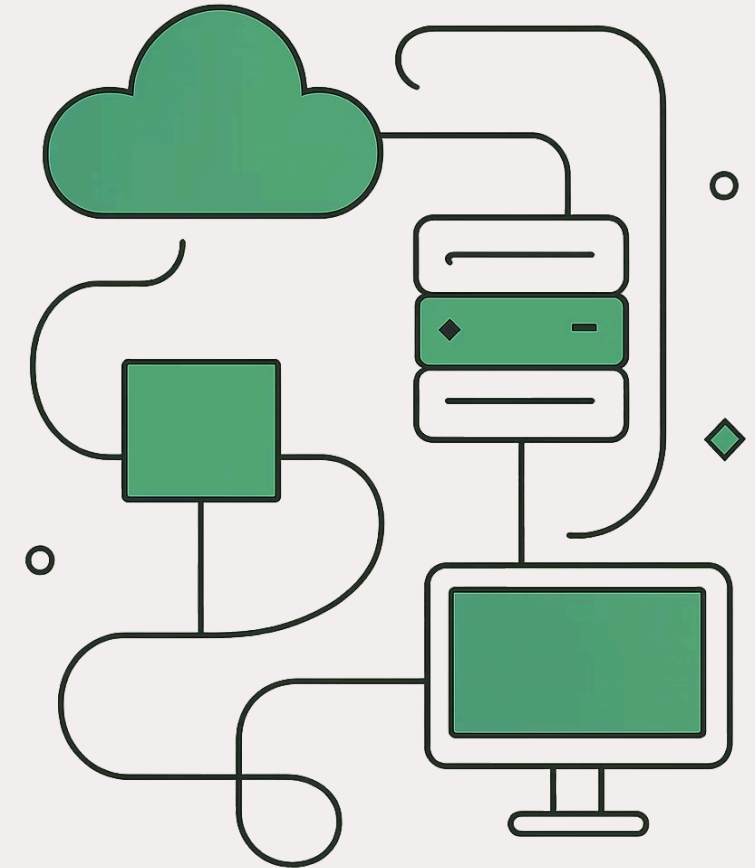
Right-sizing vSphere environments for growth

Security Hardening

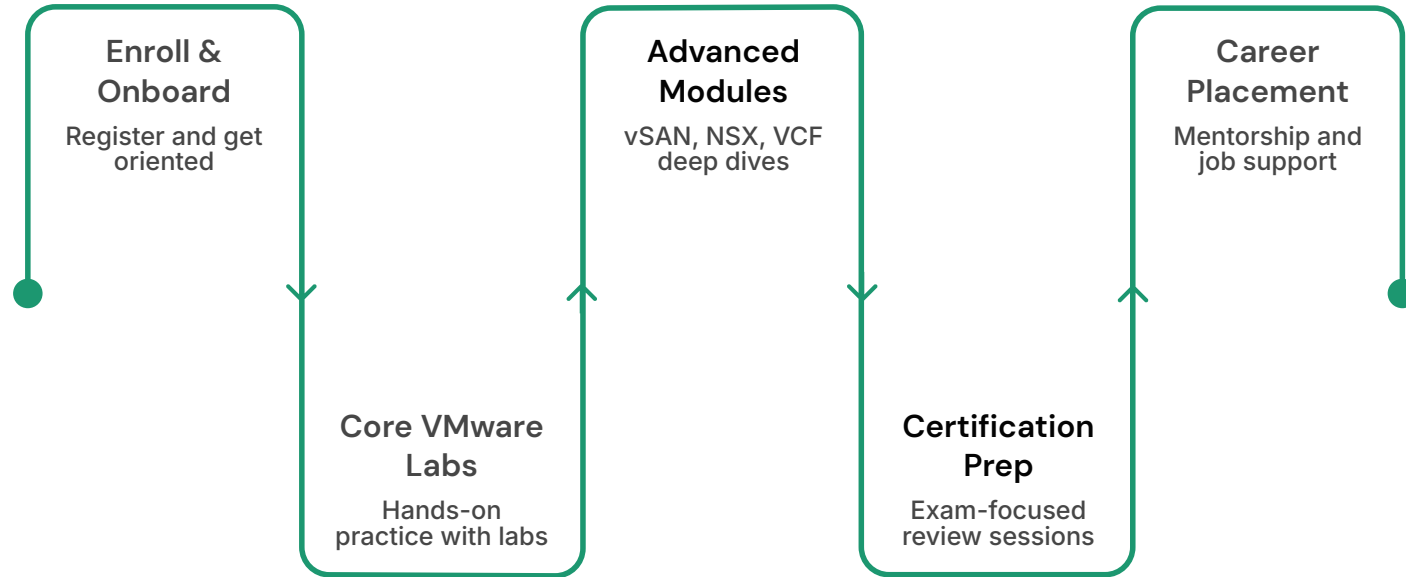
CIS benchmarks, lockdown policies, certificate management

Multi-Site Design

Stretched clusters, cross-site vMotion, private cloud architecture



Your Learning Journey



15+

Modules
Structured curriculum

100%

Live Labs
Real VMware infrastructure

40+

Lab Hours
Hands-on practice

1:1

Mentorship
Dedicated career support

Ready to Get Started?

Join SMART MATE VENTURES and build the enterprise VMware skills that top organizations demand.

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